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Credit - Drishti Ias

Open Coalition on Compliance Carbon Markets

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For Prelims: Carbon Credit Trading Scheme , Indian Carbon Market , Perform, Achieve and Trade (PAT) scheme , Carbon Border Adjustment Mechanism

For Mains: Environmental pollution and degradation, climate change, carbon markets, market-based instruments for environmental protection, industrial decarbonisation

Source: TH

Why in News?

The **Open Coalition on Compliance Carbon Markets (OCCCM)** adopted its first work plan at a meeting in Wuhan, China. The coalition, launched in May 2026 aims to align global compliance carbon markets.

- Simultaneously, India's domestic **Carbon Credit Trading Scheme (CCTS)** has become operational, with nearly 500 companies filing their first compliance reports. This has sparked a debate on **whether India should join the OCCCM to shape global carbon market rules.**

Summary

- Open Coalition on Compliance Carbon Markets (OCCCM) promotes international alignment of carbon markets, creating opportunities for India to influence global carbon accounting standards and protect exporters from carbon border taxes.
- India’s Carbon Credit Trading Scheme (CCTS) promotes industrial decarbonisation through tradable Carbon Credit Certificates. India must strengthen MRV systems, protect MSMEs and safeguard climate justice while pursuing international carbon market cooperation.

What is the Open Coalition on Compliance Carbon Markets (OCCCM)?

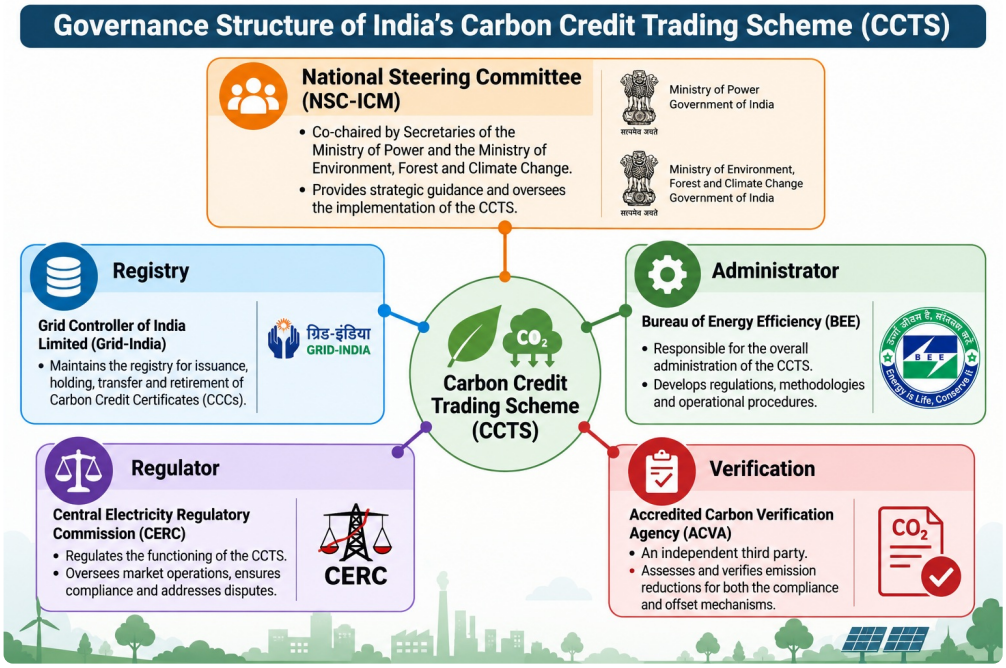
- **About:** The OCCCM is an international collaborative initiative launched to foster technical alignment,

equivalence, and interoperability among national compliance carbon markets.

- **Formation & Leadership:** Launched in May 2026, the coalition is chaired by **Brazil** , with the **European Union (EU) and China** serving as Co-Chairs.
  - Other key members include Germany, New Zealand, Canada, France, Norway, Singapore, Türkiye, and the United Kingdom.
  - Observers include **UNDP** , **UNEP** , the **World Bank's** Partnership for Market Implementation, and other international organisations.
- **Core Mandate:** It functions as a platform for exchanging knowledge on domestic compliance carbon markets, aiming to align how the world's carbon pricing architectures communicate with each other.
- **Key Areas of Focus:**
  - **Carbon Pricing Mechanisms:** Sharing strategies on emissions trading systems (ETS) and carbon taxes.
  - **MRV Systems:** Standardizing Monitoring, Reporting, and Verification (MRV) frameworks.
  - **Carbon Accounting:** Developing common methodologies for measuring emissions.
  - **High-Integrity Offsets:** Establishing rules for the potential cross-border use of carbon offsets.
- **Non-Binding Nature:** Crucially, the OCCCM **does not impose emission-reduction** targets or mandate specific domestic policies; it is primarily designed for technical alignment and interoperability.

How Does India's Carbon Credit Trading Scheme (CCTS) Operate?

- **Legislative Backing** : The CCTS is a market-based mechanism under the **Indian Carbon Market (ICM),** enabled by the **Energy Conservation (Amendment) Act, 2022,** to reduce greenhouse gas emissions and support **India's NDCs and 2070 Net-Zero target.**
  - Unlike the **Perform, Achieve and Trade (PAT) scheme,** which incentivised energy efficiency through **Energy Saving Certificates (ESCerts)** , the CCTS focuses on reducing GEI through sector-specific targets.
- **Sectors Covered:** CCTS initially covers major emission-intensive sectors such as aluminium, cement, chlor-alkali, pulp and paper, iron and steel, fertilizer, petroleum refineries, petrochemicals, and textiles.
  - Although the thermal power sector contributes a significant share of India's emissions, thermal power plants have not yet been transitioned into the CCTS compliance mechanism.
- **Baseline-and-Credit Mechanism:** Unlike EU's absolute "cap-and-trade" systems, **India utilizes a GHG emission-intensity approach.** It permits industrial output growth while mandating a reduction in emissions per unit of product.
- **Mechanisms under CCTS:**
  - **Compliance Mechanism: Obligated entities (** industries in specified sectors) are given binding GHG intensity targets. Overachievers earn tradable **Carbon Credit Certificates (CCCs)** , while underachievers must purchase them or face penalties.
  - **Offset Mechanism:** Non-obligated entities like renewable energy producers and others can voluntarily register mitigation projects under the offset mechanism to earn and sell CCCs.



What is the Need for India to Join OCCCM?

- **Shaping the Rules from the Inside:** OCCCM's governance structure (high-level meetings, thematic working groups) can dictate the future standards of global carbon accounting.
  - The geopolitical risk of being an outsider is that India may eventually be forced to **comply with MRV standards written entirely by developed nations.**
- **Proliferation of Carbon Pricing:** Over 50 countries currently utilize carbon pricing. It minimizes the economy-wide cost of mitigation and generates critical fiscal revenues for climate adaptation.
  - Joining the coalition early positions India at the forefront of this inevitable global economic shift.
- **Market Interoperability & Revenue Generation:** A long-term strategic outcome of the OCCCM is linking domestic trading systems.
  - Linking India's CCTS to international markets could allow Indian green projects to earn higher revenues by selling credits to jurisdictions with more stringent, higher-priced mitigation targets.
- **Mitigating the Threat of "Climate Clubs" & CBAM:** As UNFCCC negotiations face continuous logjams, plurilateral "climate clubs" are re-emerging.
  - The EU's **Carbon Border Adjustment Mechanism (CBAM)** converts carbon intensity directly into an international trade cost.
  - While the OCCCM does not explicitly mention CBAM, standardizing MRV and carbon accounting through the coalition is the most effective preparation for Indian exporters facing border carbon taxes.

What are the Potential Geopolitical Risks and Challenges of India Joining OCCCM?

- **Erosion of CBDR-RC:** There is a persistent concern among developing nations that global carbon market integration might blur the lines of **Common But Differentiated Responsibilities and Respective Capabilities (CBDR-RC)** , placing undue burdens on emerging economies.
- **Protectionism in Disguise:** Alignment forums can inadvertently become echo chambers for the **Global North** to legitimize carbon border taxes, which act as disguised trade barriers against the Global South.
- **Domestic Policy Autonomy:** Prematurely linking India's nascent CCTS with highly mature markets

(like the EU ETS) could restrict domestic policy flexibility, potentially harming **Micro, Small, and Medium Enterprises (MSMEs)** through overly stringent MRV compliance costs.

- **Integrity Risks:** Without robust baselines, carbon markets face risks such as "leakage" (shifting emissions to other regions) or generating non-additional credits, which undermine genuine climate action.

What Measures can India Take to Strengthen its ICM?

- **Robust Institutional Capacity (MRV):** The BEE must ensure that domestic **Accredited Carbon Verification Agencies (ACVAs)** operate at global standards. The credibility of an Indian CCC depends entirely on the rigor of its MRV system.
- **Ensure a Just Transition:** Implement complementary socio-economic policies and financial subsidies to protect vulnerable workers and MSMEs in emission-intensive sectors (like coal and traditional textiles) from sudden compliance shocks.
- **Phased Transition to Absolute Caps:** As the Indian economy matures, the CCTS should structurally transition from " **emission-intensity" targets to an absolute "Cap and Trade" model** to align with true Net-Zero trajectories.
- **Integrating Climate and Trade Policy:** The Ministry of Commerce and the MoEFCC must work in tandem. Export-heavy sectors (like steel and aluminium) must be prioritized for decarbonization subsidies to shield them from CBAM.
- **Capacity Building for MSMEs:** The government must provide financial and technical assistance to MSMEs to help them adopt carbon accounting software (like Carbon-OS) and integrate into the compliance market without facing existential financial burdens.

Conclusion

India’s Carbon Credit Trading Scheme (CCTS) marks a major step towards a low-carbon economy. Joining the OCCCM could help India promote equitable carbon accounting, protect exporters from carbon border taxes and balance climate action with economic growth.

> ***Drishti Mains Question:***
>
> How can India participate in international carbon market cooperation without compromising the principle of Common But Differentiated Responsibilities and Respective Capabilities (CBDR-RC)?

Frequently Asked Questions (FAQs)

1. What is the Open Coalition on Compliance Carbon Markets (OCCCM)?
The OCCCM is an international collaborative initiative launched in May 2026 to promote technical alignment, interoperability and knowledge-sharing among national compliance carbon markets. It is a non-binding platform.

2. What is the Carbon Credit Trading Scheme (CCTS), and what is its legal basis?
The CCTS is a market-based mechanism under the Indian Carbon Market, enabled by the Energy Conservation (Amendment) Act, 2022. It uses tradable Carbon Credit Certificates (CCCs) to incentivise greenhouse gas emission reductions.

3. How does India's baseline-and-credit model differ from the cap-and-trade model?
India's baseline-and-credit approach rewards entities that outperform prescribed emission-intensity targets. A traditional cap-and-trade system sets an overall emissions cap and allows regulated entities to trade emission allowances.

4. What is the significance of MRV in carbon markets?
Monitoring, Reporting and Verification (MRV) ensures that emissions reductions are accurately measured, transparently reported and independently verified. Robust MRV improves the credibility of carbon credits and reduces the risk of greenwashing.

5. How can joining the OCCCM benefit India, and what challenges must it address?
Participation could help India shape international carbon accounting standards, improve market interoperability and prepare exporters for CBAM-related requirements. However, India must safeguard CBDR-RC, domestic policy autonomy and MSME competitiveness while strengthening carbon market integrity.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/WOmq38KELxw]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Consider the following statements (2023)

Statement—I: Carbon markets are likely to be one of the most widespread tools in the fight against climate change.

Statement—II: Carbon markets transfer resources from the private sector to the State.

Which one of the following is correct in respect of the above statements?

(a) Both Statement—I and Statement—II are correct and Statement—II is the correct explanation for Statement—I

(b) Both Statement—I and Statement—II are correct and Statement—II is not the correct explanation for Statement—I

(c) Statement—I is correct but Statement—II is incorrect

(d) Statement—I is incorrect but Statement—II is correct

Ans: B

Q. The concept of carbon credit originated from which one of the following? (2009)

(a) Earth Summit, Rio de Janeiro

(b) Kyoto Protocol

(c) Montreal Protocol

(d) G-8 Summit, Heiligendamm

Ans: B

China’s Export Controls: Implications for India

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For Prelims: Tunnel Boring Machines , Gallium and Germanium , Graphite , Antimony , NdFeB permanent magnets , Production Linked Incentive

For Mains: India-China relations, Economic diplomacy, Strategic competition and geopolitical coercion, Weaponisation of interdependence and the use of trade restrictions as geopolitical leverage.

Source:TH

Why in News?

China announced new export controls on critical heavy machinery, including **Tunnel Boring Machines (TBMs)** , walking excavators and plateau-type wet shotcrete machines. The restrictions will take effect on 18th November 2026, introducing a dual-use export licensing regime on national security grounds.

- The move highlights **China's expanding use of supply-chain controls ,** extending beyond critical minerals to strategic industrial equipment with potential implications for India's infrastructure development and supply-chain security.

Summary

- China's export controls on critical machinery and minerals highlight the weaponisation of global supply chains, threatening India's infrastructure development, defence preparedness, pharmaceutical sector and clean energy transition.
- India must strengthen supply-chain resilience by accelerating indigenous manufacturing, expanding domestic critical mineral processing, attracting investment through the China Plus One strategy and promoting international partnerships and recycling.

How has China's Export Control Strategy Evolved Historically?

- **Export Control Law (2020):** Moving away from informal bans, China enacted a comprehensive **Export Control Law in late 2020** . This unified scattered trade rules **under a single "national security" mandate,** providing a statutory foundation to **legally throttle exports and target foreign entities.**
- **Semiconductor & Battery Minerals (2023):** In July 2023, China imposed strict licensing requirements on **Gallium and Germanium** , elements critical for advanced microchips and radar systems.
  - In October 2023, these controls were extended to **Graphite** , an indispensable material for **Electric Vehicle (EV) battery** anodes.
- **Locking the "Know-How" (Late 2023):** Realizing that competitor nations were attempting to bypass mineral embargoes, China revised its export catalogue to **outright prohibit the outward transfer of technology** used for rare earth extraction, separation, and magnet production, cementing its midstream monopoly.
- **Munitions and Escalation (2024–2025):** By late 2024, China restricted exports of **Antimony** , a mineral essential for armor-piercing munitions and solar technology, and later banned shipments to U.S. military end-users.
  - Throughout 2025, licensing was expanded to include medium and heavy rare earths, targeting **global defense and clean energy supply chains.**
- **Supply-Chain Security Regulations (2026):** China introduced State Council Orders 834 and 835, strengthening its legal framework for protecting industrial supply chains and countering foreign extraterritorial restrictions.
  - These measures could expose companies operating across Chinese and Western markets to overlapping or conflicting compliance requirements.
- **Advanced Technology Controls (2026)** : China proposed export restrictions on 40 technologies,

including 23 **artificial intelligence (AI),** 12 photovoltaic and five biotechnology technologies.

- These cover advanced AI models, semiconductor design technologies, polysilicon production and next-generation solar-cell manufacturing, raising concerns over international technology access and supply-chain dependencies.

- **Broadening to Industrial Goods (2026):** In October 2026, Beijing expanded its playbook to heavy manufacturing, implementing licensing requirements on TBMs, dredger components, and high-altitude excavators, directly targeting the infrastructure capabilities of its competitors.
  - Currently, Chinese manufacturers account for roughly 70% of the world's TBM production.

Strategic Objectives Behind the Curbs

- **Asymmetric Retaliation:** The export controls serve as a geopolitical counterweight to Western restrictions on advanced semiconductors and **Extreme Ultraviolet (EUV) lithography systems** .
  - While the US and EU leverage high-tech software and semiconductor chokepoints, China leverages physical hardware and raw input constraints.
- **Leveraging the Midstream Bottleneck:** China currently accounts for roughly 70% of global rare-earth refining capacity.
  - By banning the export of processing technology alongside the raw materials, China prevents other countries from easily setting up competing refineries, thus **solidifying its monopoly.**
  - China solidifies its manufacturing chokehold, preventing competitor nations from easily establishing parallel supply chains.
- **Economic Coercion and Relocation** By creating artificial material scarcity abroad, Beijing incentivizes downstream foreign manufacturers (like EV battery makers and electronics companies) to relocate their production facilities inside China to guarantee supply access.
- **The "Geography of Manufacturing" Vulnerability:** The curbs enforce the reality that possessing Western Intellectual Property is insufficient if the manufacturing plant is physically located in China.
  - **The Herrenknecht Precedent:** Since 2024, Chinese customs have repeatedly blocked the export of TBMs destined for Indian infrastructure projects. Even though these machines were manufactured by Herrenknecht, a renowned German company, they were built at its facility in China, making them subject to Beijing's political leverage.

What are the Implications of China's Export Control on India?

India’s heavy reliance on Chinese imports exposes multiple sectors to potential economic coercion. **India’s exports to China** rose by 40.7%, while the bilateral trade deficit widened to **USD 105.93 billion during January–September 2026.**

- **Infrastructure Stalling and Mega-Projects:** India utilizes massive TBMs weighing around 3,000 tonnes for flagship projects like the **Mumbai-Ahmedabad High-Speed Rail Corridor (Bullet Train)** and extensive underground metro networks across metropolitan cities.
  - Licensing hurdles in China will likely cause severe project delays and cost overruns.
  - The curbs on walking excavators and plateau-type wet shotcrete machines (used for applying concrete on uneven terrain) directly impact **India's strategic tunnelling projects in high-altitude border regions** (e.g., the Himalayas), which are critical for swift military mobilization.
- **Pharmaceutical Vulnerability (APIs):** India is known as the "Pharmacy of the World," yet it is

critically vulnerable upstream. India imports approximately 70% of its **Active Pharmaceutical Ingredients (APIs) from China.**

  - For certain essential antibiotics (like penicillin and cephalosporin) and generic drug building blocks, dependence approaches 80-90%. Any weaponization of API exports by **China could trigger severe domestic healthcare crises.**

- **Green Energy Transition & Electronics:** India's ambitious solar energy and Electric Vehicle (EV) targets are highly vulnerable to disruptions.
  - India depends on China for over 75% of electronic components and over 80% for solar panels, EV batteries, and laptops.
  - Disruptions in graphite, antimony, and **NdFeB permanent magnets** directly threaten India's decarbonization goals.
- **Defense Preparedness;** Modern defense platforms are heavily reliant on critical minerals. Advanced military radars **require gallium, while armor-piercing munitions and night-vision goggles require antimony** .
  - Artificial shortages threaten India's military modernization and indigenous defense manufacturing.
- **Retaliatory Trade Dynamics:** The restrictions reflect a highly combative, tit-for-tat dynamic in bilateral trade. To counter Chinese dumping, India launched 18 anti-dumping and anti-subsidy investigations in the July-September 2026 quarter alone (the highest globally) with China being the single largest target.
- **AI Access Vulnerability:** India risks being squeezed between US restrictions on AI hardware and China's control over advanced AI models. Reliance on Chinese-hosted AI services could expose government data, defence systems and citizens' personal information to security and privacy risks.

What Measures can India take to De-Risk Supply Chains?

- **National Critical Mineral Mission (NCMM):** India must rapidly implement the **NCMM** to fast-track domestic exploration, mining, and beneficiation of critical minerals, reducing upstream vulnerabilities.
- **Accelerating Indigenization (Aatmanirbhar Bharat):** The export curbs should act as a catalyst to accelerate the localization of heavy, strategic construction machinery.
  - The government must explore targeted **Production Linked Incentive (PLI)** schemes for the **heavy engineering sector** to build a purely Indian design and manufacturing ecosystem.
- **Building Midstream Capabilities:** Mining raw ore is insufficient if it must be sent to China for processing. India must deploy fiscal subsidies and public-private partnerships to build domestic **hydrometallurgical refining plants and solvent extraction complexes** to neutralize Beijing's midstream leverage.
- **Capitalising on the China Plus One Strategy:** India's revised **Press Note 3 framework** permits up to 10% non-controlling beneficial ownership from countries sharing a land border with India under the automatic route, subject to applicable conditions. It also provides 60-day fast-track approval for eligible manufacturing joint ventures in specified sectors.
  - **Global Original Equipment Manufacturers (OEMs)** are diversifying manufacturing to reduce geopolitical and supply-chain risks associated with excessive dependence on China.
  - German TBM manufacturer Herrenknecht's planned manufacturing facility in India highlights the potential to localise critical machinery production. India should facilitate technology transfer and **Foreign Direct Investment (FDI)** to strengthen domestic manufacturing and supply-chain resilience.

- **International Alliances (Friend-shoring)** India must utilize multilateral platforms to build resilient, "China-free" supply chains. Participation in the US-led **Minerals Security Partnership (MSP)** and leveraging the **Quad** (India, US, Japan, Australia) for technology transfers is critical.
- **Promoting a Circular Economy** Urban mining and recycling are crucial. The **Battery Waste Management Rules, 2022** , which mandate Extended Producer Responsibility (EPR) for end-of-life batteries, will help India recover critical metals from electronic waste, reducing import dependence.

Conclusion

China’s growing use of export controls, from rare earth minerals to advanced tunnelling machinery, highlights the weaponisation of global supply chains. For India, these restrictions threaten critical infrastructure projects but also underscore the need to accelerate indigenisation, diversify imports and strengthen domestic manufacturing to build long-term industrial resilience.

> ***Drishti Mains Question:***
>
> The weaponisation of global supply chains is transforming economic interdependence into a tool of geopolitical coercion. Examine its implications for India's strategic autonomy.

Frequently Asked Questions (FAQs)

1. What are China's new export controls on critical machinery?
Announced on October 19, 2026, the restrictions cover equipment such as Tunnel Boring Machines, walking excavators and plateau-type wet shotcrete machines. They introduce export licensing requirements effective November 18, 2026, according to the supplied article.

2. Why are Tunnel Boring Machines strategically important for India?
TBMs are essential for metro networks, high-speed rail and underground infrastructure. Advanced tunnelling equipment also supports strategic border connectivity and military mobilisation in difficult Himalayan terrain.

3. What is the National Critical Mineral Mission (NCMM)?
The NCMM aims to strengthen India's critical mineral value chain through domestic exploration, mining, processing, recycling and overseas asset acquisition, reducing import dependence and supporting strategic industries.

4. How can the China Plus One strategy benefit India?
It can attract foreign investment, technology transfer and manufacturing diversification as companies reduce excessive dependence on China. India needs competitive infrastructure, skilled labour and stable policies to capture these opportunities.

5. How do the Battery Waste Management Rules, 2022, support supply-chain resilience?
The rules establish Extended Producer Responsibility (EPR) for battery producers and support organised collection and recycling. Recovering valuable materials from used batteries can reduce demand for imported critical minerals and promote a circular economy.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/zMv-uDNsqCk**]

UPSC Civil Services Examination Previous Year Question (PYQ)

Prelims

Q. Belt and Road Initiative' is sometimes mentioned in the news in the context of the affairs of (2016)

(a) African Union

(b) Brazil

(c) European Union

(d) China

Ans: (d)

Mains

Q1. The China-Pakistan Economic Corridor (CPEC) is viewed as a cardinal subset of China's larger 'One Belt One Road' initiative. Give a brief description of CPEC and enumerate the reasons why India has distanced itself from the same. **(2018)**

Q2. "China is using its economic relations and positive trade surplus as tools to develop potential military power status in Asia". In the light of this statement, discuss its impact on India as her neighbour. **(2017)**

India to Host 61st International Committee of Legal Metrology Meeting

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Source: PIB

India will host the **61st Meeting of the International Committee of Legal Metrology (CIML)** for the **first time in New Delhi,** with participation from over 100 international delegates and legal metrology experts.

- **Organiser:** The Department of Consumer Affairs, in collaboration with the International Organization of Legal Metrology (OIML).
- **Focus Areas:** Development and revision of international recommendations, technical activities, **OIML Certification System (OIML-CS),** capacity building and harmonisation of legal metrology standards.
- **Significance for India:** Provides an opportunity to showcase India's legal metrology reforms, including simplified compliance procedures, trust-based regulation and **Ease of Doing Business** . Harmonisation of measurement standards can **reduce technical barriers to trade, minimise duplication in testing and certification** , improve export competitiveness and strengthen consumer protection.

International Organization of Legal Metrology (OIML)

- **About:** An intergovernmental organisation **established in 1955** to promote the global harmonisation of legal metrology procedures.
  - **Legal Metrology:** Application of legal requirements to measurements and measuring instruments used in **trade, industry, healthcare, safety and environmental protection** .
- **Functions** : Develops international recommendations, standards and technical documents to ensure reliable and consistent measurements across countries.
- **International Committee of Legal Metrology (CIML):** The principal decision-making body guiding the technical and administrative activities of OIML.
  - The **60th CIML Meeting (2025) was held in Paris, France.**
- **OIML Certification System (OIML-CS):** Facilitates international acceptance of type evaluation and certification of measuring instruments, helping reduce duplication in testing and certification.

Read More: **India Authorised to Issue OIML Certificates**

Nationwide Protests and Unrest in France

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Source: TH

France witnessed nationwide demonstrations against **rising living costs, deteriorating public services and poor conditions in schools** , with incidents of violent clashes, looting and arson reported in several cities.

- **Location:** Located primarily in **Western Europe** . with overseas regions and territories across the **Atlantic , Pacific and Indian Oceans.**
- **Capital: Paris** , situated on the banks of the **Seine River** .
- **Bordering Countries:** Metropolitan France shares land borders with **eight countries** : Belgium, Luxembourg, Germany, Switzerland, Italy, Monaco, Spain and Andorra.
- **Geographical Features:**
  - **Mountain Ranges:** The **Alps** lie in southeastern France, while the **Pyrenees** form much of its border with Spain.
  - **Rivers:** The Seine flows through Paris; the Loire is **France's longest river** , while the **Rhône and Garonne** are other major rivers.
  - **Coastline:** France has coastlines along the **Atlantic Ocean, English Channel and Mediterranean Sea** .
- **Important Cities:** Paris (**capital** , on the Seine River), Marseille (**major Mediterranean port**), Lyon (**confluence of Rhône and Saône rivers**), Bordeaux (on the Garonne River) and Limoges (**central-western France, where the historic town hall was damaged by fire during protests**).
- **Political System: Semi-presidential republic** , with an elected President and a government headed by the Prime Minister.
- **International Groupings:** France is a member of the **European Union (EU), Eurozone, Schengen Area, NATO and G7** .It is also one of the five permanent members of the **United Nations Security Council (UNSC)** with veto power.



Read more: New Frontiers of India-France Relations